

Wizard Duel Report Summary

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Introduction

Magic Missile Wizard Duel is a strategic 2D PVE game where the player battles with an enemy AI in a wizard duel and casts various spells. The game utilizes physics-based calculations as the players are required to find the best trajectories for their spells so that they hit and damage their opponents. This final scenario report encompasses the overall UI, main menu navigation, and gameplay flow of the Magic Missile Wizard Duel video game that was created by our group and inspired by group 3 of the spring 2024 semester.

User Interface / Main Menu Navigation

When the player first launches the game, they will encounter the game's main menu. This menu includes the buttons **Play**, **Spellbook**, **Host**, and **Join**.

Selecting **Play** immediately starts a match between the user and an enemy NPC. The user will have the ability to cast any spells previously selected by them in the spellbook in the main menu before starting the game.

Selecting **Spellbook** would open a menu that allows the user to select any available spell to include in their spell lineup/spell hotbar. These are the spells the user will take into battle upon starting a match.

Selecting **Host** would let the user create a multiplayer lobby and play against another user. Selecting **Join** on the other hand would allow the user to join an already existing lobby and play a match against the lobby's host.

As of this final scenario, we have the Play button fully implemented and working. The other buttons (Spellbook, Host, and Join) would work as previously described.

Gameplay Flow

When the game starts, the energy bars for both player and enemy wizards begin to fill. They consume built up energy to cast spells given they have enough. The wizards can launch attack spells such as the **Charge Spell** and **Ground Attack spell**, defend themselves with the **Wall Spell**, and heal themselves with the **Heal Spell**. The player controls their wizard with the mouse and keyboard, using A and D keys to move left or right respectively, 1 2 3 and 4 keys to switch between spells, and the left mouse button to cast the selected spell. Spells that require manual aiming such as the charge and wall spells utilize the mouse pointer on screen as the direction of the cast.

The game ends when either the player or enemy wizard's health drops to 0 and is knocked out of the map, after which the game will switch to the Victory or Defeat endgame scenes depending on which wizard died. From there the player can choose to return to the main menu or start a new match.

Background

The idea and concept of this videogame comes from the Wizard Duel: Development Project Design Report, by Annie You, Emily Mendoza, Kevin Cordero, and Nandana Sheri, from Spring 2024 (Group 3), and implemented by Alan Chen, Ferdinand Lim, Kareem Muftee and Xavier Hale from Spring 2026 (Group 22). This implementation utilizes the Unity game engine as a platform and is its stand-alone adaptation prototype of the game by the design team. This implementation does not follow every design detail in the design report due to design changes that would make for better gameplay, experience, as well as original designs requiring systems that would not be able to be handled within the short 3 month development period of this project.

Figure 1- Final Game Design

